Topdressing Biochar Compost Mixtures and Biological Control Organism Applications Suppress Foliar Pathogens in Creeping Bentgrass Fairway Turf

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Abstract

Biochar, compost, and biological control agents can suppress pathogens on their own; however, their reliability and efficacy are not as acceptable as synthetic fungicides commonly used to suppress pathogens. A multiyear field study was initiated to evaluate combinations of monthly applications of a biochar compost mixture and weekly or biweekly *Bacillus subtilis* QST713 applications for their ability to suppress foliar pathogens on a creeping bentgrass (*Agrostis stolonifera* L.) fairway and to measure their impact on strain QST713 establishment. Disease severity and turfgrass quality were measured every 14 days throughout the growing season. Populations of strain QST713 were quantified by quantitative PCR analysis on DNA extracted from foliage samples collected throughout the trial. Biochar compost mixture applications increased turfgrass quality in both years of the study and reduced dollar spot (*Clarireedia jacksonii* Salgado-Salazar) severity in 2021. Weekly strain QST713 applications reduced copper spot (*Gloeocercospora sorghi* D. C. Bain & Edgerton) severity compared with biweekly applications and the nontreated control in 2020, yet monthly biochar compost mixture with weekly strain QST713 applications completely suppressed copper spot in 2021. Populations of strain QST713 were highest in weekly treated plots, and monthly biochar compost mixture applications did not affect strain QST713 establishment. Although there was not an interaction between biochar compost mixture and strain QST713 applications, implementing both in a season-long program will benefit turfgrass health and reduce disease severity.

Keywords: biochar, compost, cultural and biological practices, fungi, turf

Plant pathogens have damaged all crops and caused losses for centuries (Agrios 2005). Although synthetic fungicides are commonly used to mitigate these losses, alternative methods of suppressing plant pathogens can help reduce reliance on synthetic fungicides. Biochar (i.e., pyrolyzed biomass) is an organic amendment that has multiple qualities that support incorporating it into alternative or wholistic disease management programs. A meta-analysis of 371 studies utilizing biochar as a soil amendment revealed increased soil nitrogen (N), phosphorus, potassium, and carbon; crop yields; soil microbial biomass; and rhizobia nodulation (Biederman and Harpole 2013). Amending biochar to the sand-based root zone of a golf course putting green increased water retention and creeping bentgrass growth compared with the nonamended root zone (Brockhoff et al. 2010; Vaughn et al. 2018). However, it is important to note that there is significant variation in the degree to which many of these soil physiochemical properties are affected. Furthermore, biochar may be beneficial in reducing damage from plant pathogens. Beckley and Roberts (2022) observed reductions in dollar spot (*Clarireedia* spp. Salgado-Salazar) from biochar incorporation on a creeping bentgrass fairway in both years of their study but reductions were also similar to synthetic fertilizer applied with and without vermicompost extract. Potting growth substrates amended with biochar suppressed gray mold (*Botrytis cinerea* Pers.) and powdery mildew (*Leveillula taurica* (Lév.) G. Arnaud) of pepper (*Capsicum annuum* L.) and tomato (*Lycopersicum esculentum* L.) (Elad et al. 2010) and early blight (*Alternaria solani* Sorauer) of tomato (Rasool et al. 2021). The mechanism behind biochar-mediated disease suppression in strawberry was an increased expression of five plant defense-related genes (Harel et al. 2012), which suggests that biochar may provide broad base-level benefits to plant health. Increased expression of plant defense-related genes has the potential for broad suppression of plant pathogens and needs further research, especially in turfgrass systems. However, caution should be used because having plants constitutively express defense mechanisms will result in costs to fitness and growth (Denancé et al. 2013; Vos et al. 2013).

Similar to biochar, biological control organisms (BCOs) are another alternative that have shown some degree of success in suppressing plant pathogens in the field (Pal and Gardener 2006). However, the adoption of BCOs by growers remains limited. Reliability of BCOs remains one of the chief limitations to their adoption. In a summary of 24 Plant Disease Management Reports spanning multiple turfgrass diseases, biofungicides (i.e., commercially formulated biological control organism products) reduced disease in only 33% of trials (Latin 2011). Lack of reliability may stem from resident microbiota outcompeting introduced BCOs. Research shows that turfgrass hosts have diverse microbial communities (Beirn et al. 2016; Crouch et al. 2017; Doherty et al. 2021) that will likely provide significant competition for applied BCOs to overcome in order to establish themselves in the environment. Considering that BCOs suppress pathogens through competition, antibiosis, predation, or plant defense activation, a BCO's population levels may be a significant factor in efficacy.

Alongside biochar and BCOs, supplemental cultural practices and nutrient management can be foundational to reducing disease severity by disrupting the disease triangle. For example, anthracnose (*Colletotrichum cereale* Manns) severity on annual bluegrass (*Poa annua* L.f. reptans (Hausskn) T. Koyama) putting greens can be reduced without fungicide applications by increasing mowing heights, increasing sand topdressing rates, and increasing rates of N (Hempfling et al. 2017). In some cases, these three practices can reduce anthracnose severity as effectively as fungicide applications. Severity of dollar spot, which is favored by low fertility and prolonged periods of leaf moisture, can be reduced through cultural practices such as lightweight rolling (Nikolai et al. 2001), N fertilizer

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applications (Golembiewski and Danneberger 1998), and plant growth regulators (Golembiewski and Danneberger 1998; Putman and Kaminski 2011). These management recommendations work to disrupt the disease triangle by reducing favorable conditions for the pathogen and by improving a plant's health. Defense capabilities are likely improved by increasing fertility as well because a plant that cannot meet its nutritional needs may be less able to defend itself. There is a fine balance to achieve here, however, as additional fertility can help in reducing dollar spot but excessive N fertility is traditionally associated with increased brown patch (*Rhizoctonia solani* Kühn) severity due to increased succulent new growth. Individual cultural practices are typically not enough to control diseases as well as synthetic fungicides do. As such, investigating all possible alternative practices to quell disease development is key to reducing reliance on synthetic fungicides.

Bacillus subtilis QST713 is a BCO that has been available in the turfgrass market for many years and has been successful in decreasing disease severity of foliar pathogens in multiple studies on golf course putting green turfgrass (Mattox et al. 2018; Roberts et al. 2019). Perhaps, by combining strain QST713 with a biochar compost mixture on an area of turfgrass with a higher level of acceptable disease levels (i.e., a golf course fairway), acceptable levels of disease control can be achieved in the absence of synthetic pesticides. The objectives of this study were to (i) assess combinations of monthly topdressing applications of a biochar compost mixture with weekly or biweekly biofungicide applications for their ability to suppress foliar diseases and (ii) monitor BCO populations in response to treatment combinations. We hypothesized that combinations would suppress foliar diseases more than the biochar compost mixture or biofungicides alone, that BCO populations would be higher in combination treatments than in biofungicide-only treatments, and that weekly BCO applications would result in higher BCO populations and improved disease suppression.

Materials and Methods

Site description

A 2-year field trial was initiated in May 2020 on a creeping bentgrass (*Agrostis stolonifera* L. '007') fairway at the Paint Branch Turfgrass Research Facility in College Park, MD. A previous study, conducted from 2016 to 2018, investigating organic soil amendments for dollar spot suppression was located at the trial area (Beckley and Roberts 2022). The trial area was mowed three times per week at a bench height of 1.25 cm with a Jacobsen Greens King IV Plus triplex reel mower (Jacobson Manufacturing, Ipswich, England, U.K.). The trial area was irrigated to prevent drought stress. Supplemental N fertility was applied as soluble urea (46-0-0) with N at a rate of 44 kg ha^{-1} on 20 April 2020 and 12 April 2021. Annual weeds were controlled preemergence with dithiopyr (Dimension 2EW; Dow AgroSciences, Indianapolis, IN, U.S.A.) with active ingredient at a rate of 0.56 kg ha^{-1} on 28 April 2020 and 12 April 2021. Annual bluegrass (*P. annua* L.f. reptans (Hausskn) T. Koyama) was controlled postemergence with amicarbazone (Xonerate 2SC; FMC Corp., Philadelphia, PA, U.S.A.) with active ingredient at a rate of 0.17 kg ha^{-1} on 28 April 2020, 13 May 2020, 27 April 2021, and 10 May 2021. Fertilizer and herbicide applications were made using a Toro Multi Pro 1750 (The Toro Company, Bloomington, MN, U.S.A.) equipped with TeeJet AIC11008 nozzles calibrated to deliver applications in H_2O at 814.9 liters ha^{-1}.

Experimental design

This trial was arranged as a split-plot randomized complete block design with a 2-by-3 factorial treatment arrangement. Treatment combinations were replicated four times. Dimensions of the main plots measured 1.8 by 6.1 m and were split into three subplots measuring 1.8 by 1.8 m, with a 0.3-m buffer between subplots. Main plot treatments in this study were a modified selection of the main plot treatments from the previous study by Beckley and Roberts (2022). The main plots receiving biochar compost mixture topdressing applications and the nontreated main plots from their study were located and used for this study. This study used the same biochar compost mixture (2-2-2 Mirimichi Green Pro Soil Enhancer; Mirimichi Green Express, Castle Hayne, NC, U.S.A.) used by Beckley and Roberts (2022). By using the same plots and biochar compost mixture, this study can take advantage of the legacy effects from the soil amendment. Applications of the biochar compost mixture in this study were made monthly from May through August each year with N at a rate of 20.4 kg ha^{-1}. Topdressing was performed using a shaker jar and a hard landscape rake to evenly distribute the application across the plot area.

Subplot treatments consisted of weekly and biweekly *B. subtilis* QST713 (Rhapsody; Bayer Crop Science, Research Triangle Park, NC, U.S.A.) and a nontreated control. In both application intervals, *B. subtilis* QST713 was applied at a rate of 0.308 kg ha^{-1} (Rhapsody at 31.8 liters ha^{-1}). Treatments were applied through a CO_2-pressurized system using an air-induction nozzle calibrated to apply 814.9 liters ha^{-1} evenly across the plot. Subplot treatments were applied at their respective intervals from 11 June to 19 August 2020 and from 2 June to 11 August 2021.

Dollar spot and copper spot (*Gloeocercospora sorghi* D. C. Bain & Edgerton) severity were evaluated by counting infection centers within each plot. Brown patch severity was evaluated through visual estimation of percent area within plots exhibiting brown patch symptoms. Turfgrass quality (i.e., color, density, and uniformity) was evaluated visually on a 1-to-9 scale, where 6 represented minimal acceptable quality and 9 represented the highest quality possible. Disease severity and turfgrass quality were evaluated every 7 to 14 days throughout the trial period each year. Area under the progress curve (AUPC) values were calculated using the trapezoidal method to produce summative ratings of disease severity and turfgrass quality in each year (Madden et al. 2007).

Monitoring strain QST713 populations

To evaluate the establishment of strain QST713 in response to treatment combinations, samples were collected in 2020 on 22 May and 19 August and in 2021 on 25 May, 22 June, 21 July, and 18 August. May sampling dates were prior to treatment applications each year to serve as a starting point, and all other sampling dates were 6 days after an application date where biweekly and weekly strain QST713 applications were made. For each subplot, samples of five turfgrass plants were collected, using flame-sterilized forceps, from four randomized locations within the plot. Senescent foliage and root tissue were removed, and samples were placed into preweighed 1.5-ml microcentrifuge tubes. Tubes were kept on ice in the field and during transit to the lab, whereupon they were reweighed and placed into a −20ºC freezer to limit community degradation. Samples were lyophilized for 48 h prior to DNA extraction to optimize yields. DNA was extracted from samples using the Qiagen DNeasy Plant Mini Kit (Qiagen, Gaithersburg, MD, U.S.A.), following the manufacturer's protocol.

Pure cultures of strain QST713 were isolated through dilution plating of the commercial product Rhapsody (Bayer Crop Science). Once a pure culture was obtained, strain QST713 cells were suspended in sterile deionized H_2O and diluted to an optical density of 1.0 (approximately 10^9 CFU ml^{-1}). DNA was extracted from cellular suspensions using the Zymo Quick-DNA Fungal/Bacterial Miniprep kit (Zymo Research Corporation, Irvine, CA, U.S.A.) following the manufacturer's instructions. Extracted DNA was quantified using a NanoDrop One spectrophotometer (Thermo Fisher Scientific, Waltham, MA, U.S.A.). DNA was then diluted to a concentration of strain QST713 DNA of 10 ng μl^{-1}. A serial dilution was performed to obtain additional strain QST713 DNA concentrations of 1, 0.1, 0.01, and 0.001 ng μl^{-1}, forming a standard curve with five steps.

Using strain QST713 primers, probe, and protocol as previously described (Mendis et al. 2018), qPCR assays were completed in the StepOnePlus Real-Time PCR System (Applied Biosystems, Foster City, CA, U.S.A.) with a final volume of 20 µl, containing 2 µl of sample DNA, PerfeCTa qPCR ToughMix ROX (Quantabio, Beverly, MA, U.S.A.), 250 nM each forward primer and reverse primer, and 150 nM TaqMan probe (Integrated DNA Technologies, Coralville, IA,

U.S.A.). Thermal conditions for qPCR assays were as follows: 95ºC for 15 min followed by 39 cycles of 95ºC for 15 s and 58ºC for 1 min.

Statistical analysis

Data from qPCR assays were analyzed using a two-way repeated measures analysis of variance (ANOVA) and AUPC values were subjected to ANOVA in SAS 9.4 (SAS Institute, Cary, NC, U.S.A.). Logarithmic transformations were applied to qPCR data to correct patterns in the residuals. For statistically significant factors, Tukey's honestly significant difference post hoc test was used to compare treatment means. Graphs were generated using 'ggplot2' in R (Wickham et al. 2016).

Results

Field data

Dollar spot developed naturally in June each year, brown patch developed naturally in July 2020, and copper spot developed naturally in August 2020 and late July 2021. Brown patch severity peaked in the nontreated control plots on 21 July 2020, with an average of 36% symptomatic plot area, and no brown patch developed in 2021. Dollar spot severity peaked in the nontreated control plots on 3 August 2020, with an average of 25 foci per plot, and on 13 July 2021, with an average of 32 foci per plot. Copper spot severity peaked in the nontreated control plots on 3 August 2020, with an average of 82 foci per plot, and on 26 July 2021, with an average of 35 foci per plot. In 2020, dollar spot severity was unaffected by treatments, in combination or alone. Topdressing applications of the biochar compost mixture reduced brown patch in 2020 and dollar spot severity in 2021, regardless of strain QST713 applications (Fig. 1A and B). Similarly in both years, plots receiving monthly biochar compost mixture topdressing had significantly higher quality than plots not receiving biochar compost mixture topdressing (Fig. 2). Turfgrass quality was never acceptable in plots receiving no topdressing treatments in either year (Supplementary Fig. S1). Weekly strain QST713 applications reduced copper spot severity by 50% compared with biweekly and nontreated control treatments in 2020, and by 95% compared with the nontreated control in 2021 (Fig. 1C), regardless of biochar compost mixture topdressing treatments.

Strain QST713 populations

Biochar compost mixture applications did not affect populations of strain QST713 in either year of the study, whereas application frequency of strain QST713 was significant across both experimental years. In 2020, populations of strain QST713 were highest in plots receiving weekly strain QST713 applications, and there was no difference between the nontreated control and biweekly strain QST713 applications (Fig. 3). By May 2021, strain QST713 was undetectable in all plots. After treatment applications resumed in 2021, strain QST713 populations were highest in plots receiving weekly strain QST713 applications and detection of strain QST713 showed increasing populations on the second and third sampling dates, before a decline on 18 August once applications had been suspended (Fig. 3).

Discussion

Applications of the biochar compost mixture proved to be more beneficial than strain QST713 in reducing disease and improving turfgrass quality. The observed benefits from biochar compost mixture applications may stem from alterations to microbial

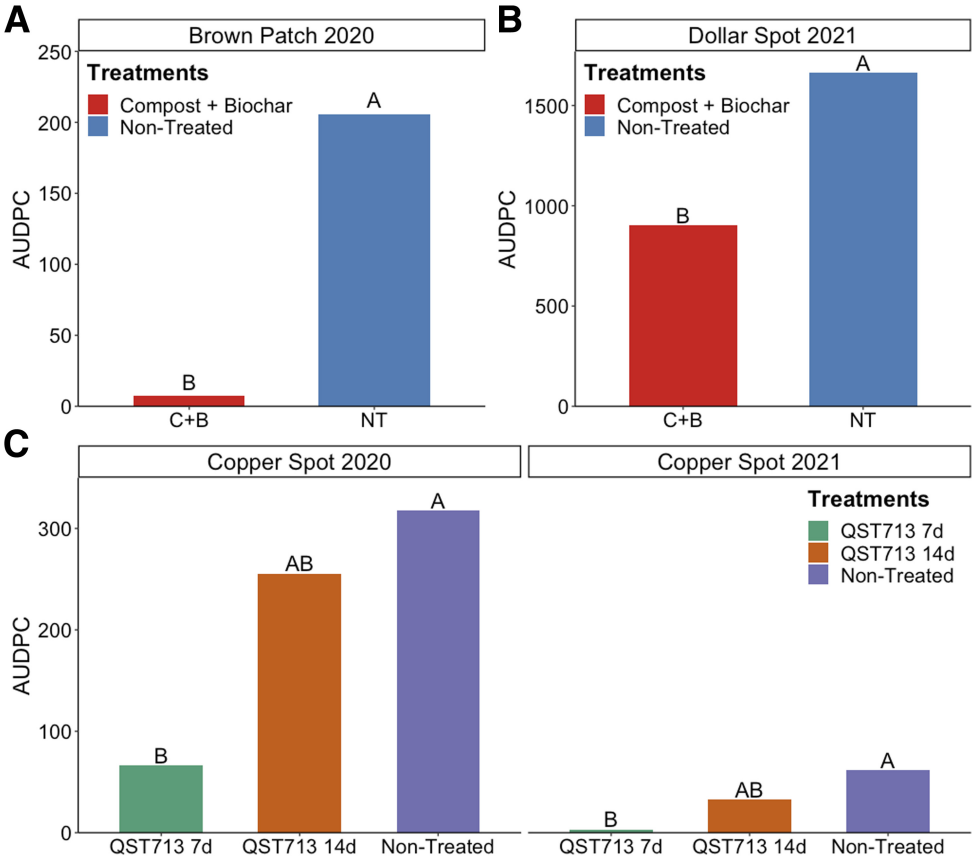


Fig. 1. Area under the disease progress curve (AUDPC) values in response to **A and B,** monthly compost + biochar topdressing applications and **C,** weekly or biweekly *Bacillus subtilis* QST713 applications. Values calculated from A, 2020 brown patch severity (percent plot area symptomatic); B, 2021 dollar spot (infection centers); and C, 2020 and 2021 copper spot (infection centers). Treatments within each quadrant sharing a letter are not significantly different according to Tukey's honestly significant difference ($P < 0.05$). C + B, QST713, and NT refer to compost + biochar, *B. subtilis* QST713, and nontreated, respectively.

communities, increased nutrient availability, or a combination of the two. Microbiota associated with N cycling have been shown to be stimulated by biochar applications (Clough et al. 2013; Xu et al. 2014), which may lead to increased N availability to the plants. However, some biochar material may decrease plant N concentration (Lentz and Ippolito 2012; O'Toole et al. 2013), while some increase it (Jones et al. 2012; Uzoma et al. 2011), and still others have no effect (Jones et al. 2012; Prendergast-Miller et al. 2011). The biochar material used in this study also contained compost; thus, perhaps more plant-available N was responsible for the observed disease reduction from biochar compost mixture treatments. In this study, biochar compost mixture-treated main plots received N at 125.6 kg ha^{-1} annually whereas nontreated main plots received N at 44 kg ha^{-1} in April. The nontreated main plots were well below the recommended annual N fertility rate of 97.81 kg ha^{-1} for creeping bentgrass fairways in Maryland (Turner 2013). Being significantly underfertilized, the nontreated main plots were likely N stressed and less capable of producing sufficient defense responses compared with the main plots receiving the biochar compost mixture (Mur et al. 2017). Furthermore, bulk soil sampling showed that long-term biochar compost mixture applications increased plant available soil micronutrients, estimated N release, and soil organic matter (data not shown), which may have also contributed to lower disease incidence and higher turfgrass quality. Further research would be needed to test these hypotheses but confirmation would explain decreased dollar spot severity, because increased N fertility has been documented to reduce severity of diseases such as dollar spot (Beckley and Roberts 2022; Golembiewski and Danneberger 1998) and increase turfgrass quality.

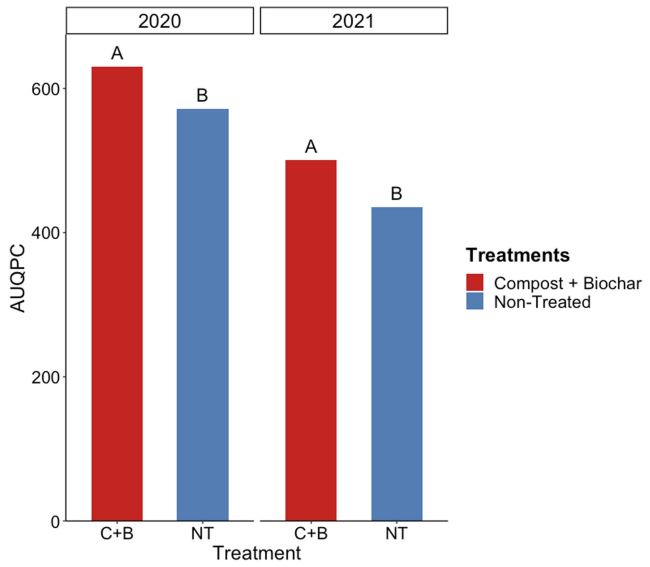
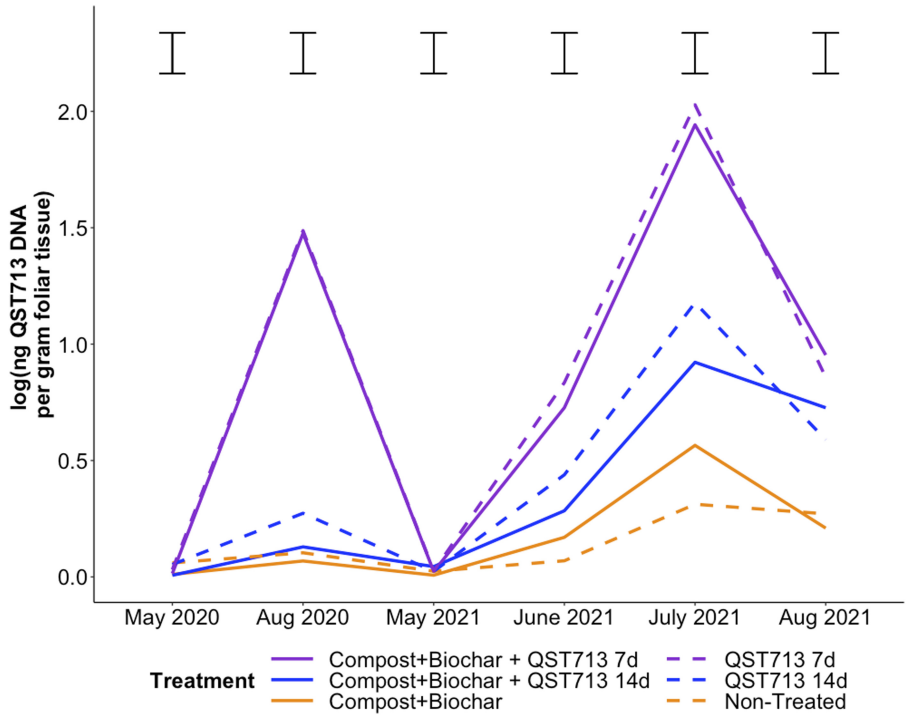


Fig. 2. Area under the quality progress curve (AUQPC) values in response to monthly compost + biochar topdressing applications. Values calculated from visual turfgrass quality ratings collected in 2020 and 2021. Treatments within each year sharing a letter are not significantly different according to Tukey's honestly significant difference ($P \leq 0.05$). C + B and NT refer to compost + biochar and nontreated, respectively.

Brown patch is often associated with higher levels of N fertilization but recent research by Butler et al. (2019) showed rates of N of 293.1 kg ha^{-1} year^{-1} or higher for 2 years were needed to intensify brown patch severity compared with the nontreated control. Neither main plot treatment exceeded the rate reported by Butler et al. (2019); however, as mentioned above, the nontreated main plots were well below the recommended annual N rate and this may lead to compromised defense response. Alternatively, the suppression of brown patch may stem from alterations to the plant microbiome in response to biochar applications. Kolton et al. (2017) observed biochar increasing microbial and functional diversity of tomato rhizosphere bacterial communities and decreasing gray mold severity. Additionally, biochar did not result in changes to plant physiological characteristics, leading the authors to believe that the alterations to microbial communities were a significant factor in the reduced gray mold severity. Plant defense mechanisms have also been observed to be elicited by biochar applications (Harel et al. 2012). Biochar applications upregulated plant defense pathways in tomato and led to a reduction in Fusarium crown rot severity (Jaiswal et al. 2020). It is important to note that, in most studies of biochar, it is incorporated into the soil or potting growth medium whereas, in our study, the material was applied monthly over the top of the turfgrass. Our findings suggest that topdressing biochar may still provide disease suppression benefits that have been previously observed with soil-

Fig. 3. Foliar concentrations of *Bacillus subtilis* QST713 DNA, measured through quantitative PCR assays, in response to combinations of monthly compost + biochar topdressing applications and weekly or biweekly *B. subtilis* QST713 applications made to a 007 creeping bentgrass fairway. Error bars represent standard errors of the means for each sampling date.



incorporated biochar, although the underlying mechanisms remain to be elucidated in biochar topdressing.

In the absence of a regular fungicide program, uncommon diseases may become more problematic. We observed an outbreak of copper spot in both years of this study. Although copper spot has been described for some time (Smith et al. 1989), little information is available regarding fungicide efficacies (Clarke et al. 2019). It is likely that fungicide programs for managing more persistent or problematic turfgrass pathogens also control copper spot. Therefore, limited data exists for fungicide efficacy compared with more common turfgrass pathogens. Regardless, copper spot is often considered a nuisance disease that is of low priority, because turfgrass readily recovers from *G. sorghi* damage. Given the results of our research, switching away from conventional management regimes with synthetic fungicides may result in an increase in epidemics of less common or nuisance diseases such as copper spot. This highlights the need to increase our understanding of how management regimes are affecting the plant microbiome which, in turn, may elucidate pathogens that could become more problematic in the absence of fungicides.

Weekly applications of strain QST713 did result in higher strain QST713 populations at every sampling date during the growing season but these populations completely declined between field seasons (i.e., August 2020 to May 2021 and August 2021). This data supports our hypothesis that weekly BCO applications will result in higher BCO populations. The higher strain QST713 populations in the weekly treatment likely stem simply from the continual addition of strain QST713 propagules before populations from the previous treatment declined. However, the higher populations did not always translate into increased disease suppression, as one might expect given BCO modes of action. Higher populations of strain QST713 coincided with increased suppression of copper spot, but dollar spot and brown patch were unaffected by strain QST713 populations. This is similar to what was previously observed in a creeping bentgrass putting green, where higher levels of *B. subtilis* QST713 did not increase suppression of Pythium root rot (Doherty and Roberts 2022). Interestingly, strain QST713 populations rapidly declined after treatments were stopped regardless of application intervals. The composition of resident microbial communities and nutrient availability are important for introduced biological organisms' ability to compete and persist in their new environment (Yang et al. 2017). Perhaps the nutrient availability in the turfgrass phyllosphere is too poor for the introduced strain QST713 to outcompete resident communities and establish a persistent population. Although the role played by a BCO's population level in its efficacy is not well understood at this time, our data indicate that weekly applications are needed to maintain higher BCO populations.

Findings from this research suggest that implementing both products in a season-long program is advisable for improved turfgrass quality in the absence of synthetic chemicals or fertilizers. Biochar compost mixture applications reduced disease severity that strain QST713 did not, and vice versa, in addition to providing increased turfgrass quality. Populations of strain QST713 were not affected by biochar compost mixture applications, while weekly applications of strain QST713 resulted in higher populations than biweekly applications. Future research efforts are needed to understand the plant–pathogen–BCO interaction web to facilitate BCO usage recommendations. Perhaps future research efforts can utilize multiple "-omics" and molecular tools to understand pathogen–BCO population dynamics and interactions, and the underlying mechanism behind suppression. Additionally, deeper investigations into any possible correlations between a BCO's ability to suppress a pathogen and the aggressiveness of said pathogen can aid in more proper deployment of BCOs in field settings. Finally, characterizing whether the biochar compost mixtures elicited broad-spectrum defense pathways in the turfgrass or whether they altered microbial activity to suppress pathogens can aid our ability to implement additional cultural practices or applications that will complement the biochar compost mixture's effects. It is crucial to continue exploring alternative disease control measures to reduce the turfgrass industry's reliance on synthetic fungicides.

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